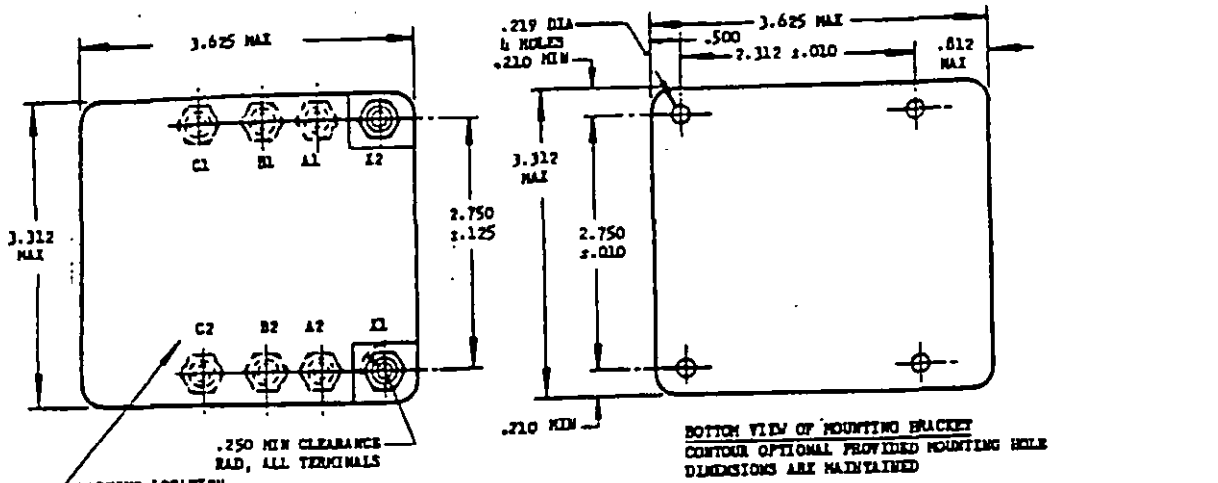
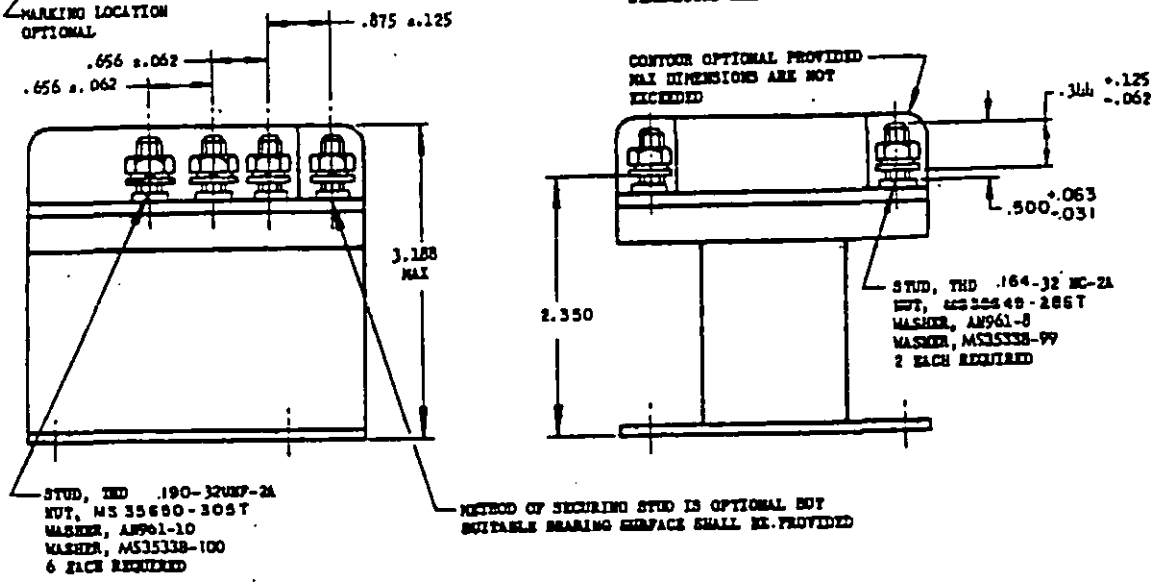
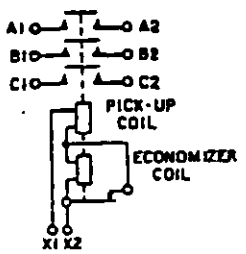
FED. SUP CLASS
5945







Circuit diagram

APPROVED 26 SEPT 1957 REVISED Ⓖ 17 JUNE 68

Ⓖ ENTIRE STANDARD REVISED

P.A	Interested	TITLE	MILITARY STANDARD
AF-85 Other Cust NAVY-AS	-	RELAY ELECTROMAGNETIC, 50 AMPERES, 3 PST, N.O. TYPE II, NON - HERMETICALLY SEALED	MS 24193
Procurement Specification MIL-R-6106		SUPERSEDES AN 3341	PAGE 1 OF 5

5945-0772

User activities: Army - Navy - Air Force -

Review activities: Army - AS, EC Navy - Air Force - 99, 11

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Inches	mm	Inches	mm	Inches	mm
.010	.25	.210	5.33	.875	22.23
.031	.79	.219	5.56	2.350	59.69
.062	1.57	.250	6.35	2.750	69.85
.063	1.60	.344	8.74	3.188	80.98
.125	3.18	.500	12.70	3.312	84.12
.164	4.17	.656	16.66	3.625	92.08
.190	4.83	.812	20.62		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.031 (0.79 mm).
4. Additional flat washer may be used for terminal seat.
5. Terminal covers and barriers required at power terminals.
6. Part number MS24193-D1 replaces number AN3341-1.
7. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
8. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

Dash number MS 24193 -	Type	Coil	Terminal type	Mounting or mating socket	Max weight in pounds
D1	II	dc	Stud	Plate	1.6

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Review activities: Army -
Navy - AS, EC
Air Force - 99, 11

User activities: Army -
Navy -
Air Force -

TABLE II. Operating Characteristics.

MS part no. MS24193-	Coil data												Time (milliseconds-maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/	Operate 3/	Release 4/	Bounce 5/			
		Volts 1/	Freq. Hz	Resn ±10% 25°C	Volts	Amperes	Normal 2/	High temp test	Cont. current test					Main		Aux	
														NO	NC	NO	NC
				In-rush / Sty state		In-rush / Sty state											
D1	X1, X2	28	dc	13.5 / 71.5	29	2.4 / .45	18	21	22.5	7.0	1.5	20	15	4			

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ Bounce time after life test 8 ms.

P.A
AF - 85
Other Cust
NAVY-AS

Procurement Specification
MIL-R-6106

Interchangeable

SUPERSEDES
AN 3014-10-23T14:50Z

TITLE
RELAY ELECTROMAGNETIC, 50 AMPERES, 3 PST, N.O. TYPE II, NON - HERMETICALLY SEALED

MILITARY STANDARD
MS24193

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Review activities: Army -
Navy - AS, EC
Air Force - 99, 11

User activities: Army -
Navy -
Air Force -

DD FORM 1 MAY 73 672

P.A AF-85 Other Cust NAVY-AS

Procurement Specification MIL-R-6106

Superseded: AN3341

TITLE RELAY ELECTROMAGNETIC, 50 AMPERES, 3 PST, N.O. TYPE II, NON - HERMETICALLY SEALED

MILITARY STANDARD

MS24193

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TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	50				50				50				
Inductive	10	50				50				50				
Inductive														
Motor	50	50				50				50				
Lamp														
Transfer load														2/
Mechanical life reduced current	100	12.5				12.5				12.5				
Intermediate current	50	5				5				5				

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Review activities: Army -
Navy - AS, EC
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User activities: Army -
Navy -
Air Force -

P.A
AF-85
Other Cust
NAVY-AS

Procurement Specification
MIL-R-6106

Interchangeable

SUPERSEDES:
AN3341

TITLE
RELAY ELECTROMAGNETIC, 50 AMPERES, 3 PST, N.O. TYPE II, NON - HERMETICALLY SEALED

MILITARY STANDARD
MS24193

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DD FORM 1 MAY 73 672 (Coordinated previous editions of this form are obsolete.)

Environmental characteristics

Temperature range	-55°C to +71°C
Max altitude rating	50,000 ft
Shock G-level	25 g's
Duration	6 - 9 ms
Max duration contact opening	2 ms

Vibration - Sinusoidal

5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1500 Hz
.08 DA	.06 DA	2 g's	2 g's	

Vibration - Random	
Applicable specification	N/A
Power spectral density	N/A
RMS G-min	N/A
Frequency range	N/A
Curve	N/A
High shock	N/A
Acceleration	10 g's

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms
Dielectric strength (sea level)	2-5 seconds

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250 V rms	N/A	1,000 V rms	N/A
Aux contacts	1,250 V rms	N/A	1,000 V rms	N/A
All other points	1,250 V rms	1,500 V rms	1,000 V rms	1,125 V rms

Dielectric strength (altitude) 1 minute

	28 V dc	115 V ac
Coil to case	500 V rms	N/A
Aux contacts	500 V rms	N/A
All other points	500 V rms	500 V rms

Max contact drop initial		0.150 volt
After life test		0.175 volt
Overload current	(N.O.)	400 amperes
Rupture current	(N.O.)	500 amperes
Duty rating		Continuous
RFI specification		MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Qualification By Similarity.

Part Number	Loads						Dynamics 1/			Environmental		
	Type 1			Type 1 ER								
	A	B	C	D	E	F	A	X	Y	A	B	C
MS24193-D1	4						2			4		

1/ All units must be tested.

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